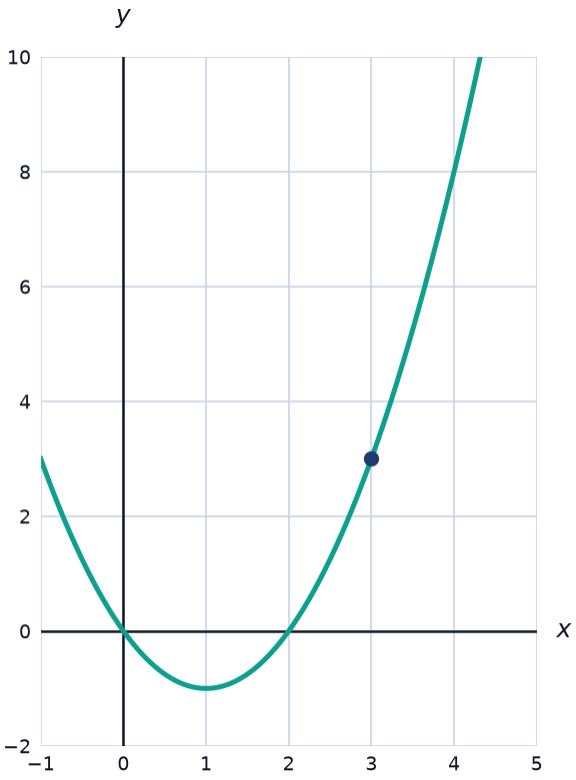
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Tangent lines**

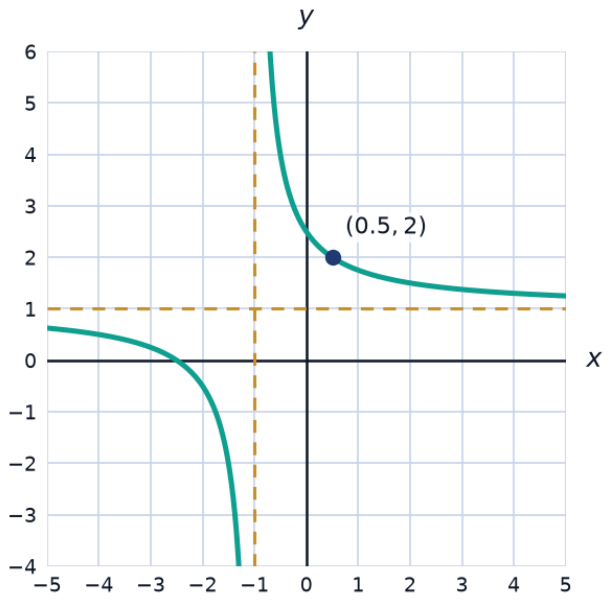
The graph of $f(x) = x^2 - 2x$ is shown. Find the equation of the tangent line at $x = 3$ (the marked point).



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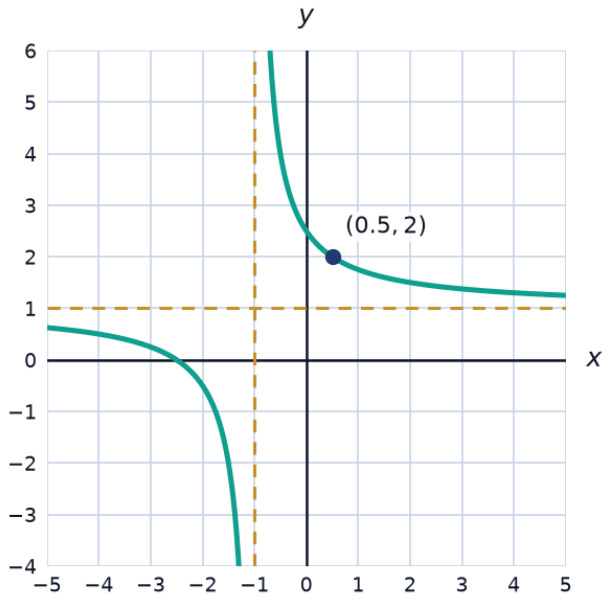
2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate $\lim_{x \to 2} (3x^2 - 4x + 1)$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

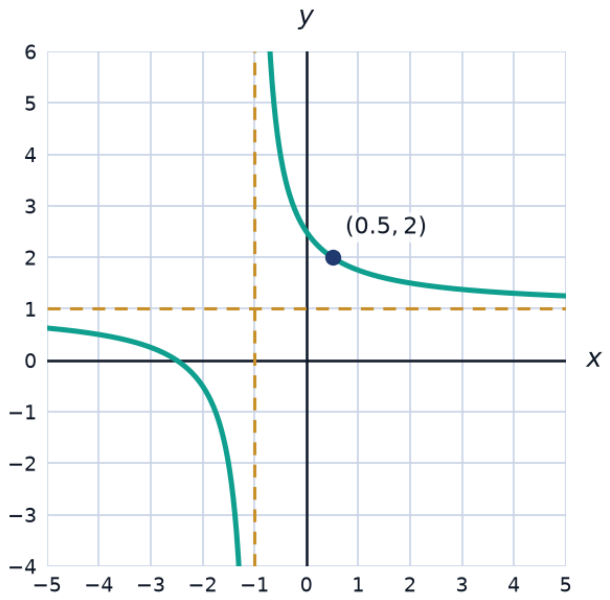
Evaluate $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$.



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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate $\lim_{x \to \infty} \frac{3x^2 + 2x}{x^2 - 5}$.

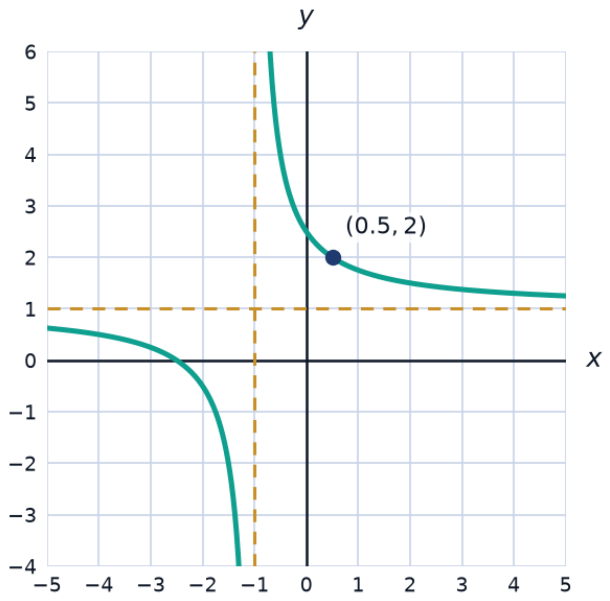


5 [Foundational] • Target: Evaluating limits by direct substitution

Evaluate $\lim_{x \to 2} (3x^2 - 4x + 1)$.

6 [Foundational] • Target: Evaluating limits by direct substitution

Evaluate $\lim_{x \to -1} (x^3 + 2x - 5)$.



7 [Foundational] • Target: Evaluating limits by direct substitution

Evaluate $\lim_{x \to 0} \frac{5x + 7}{x + 3}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Evaluating limits by direct substitution**

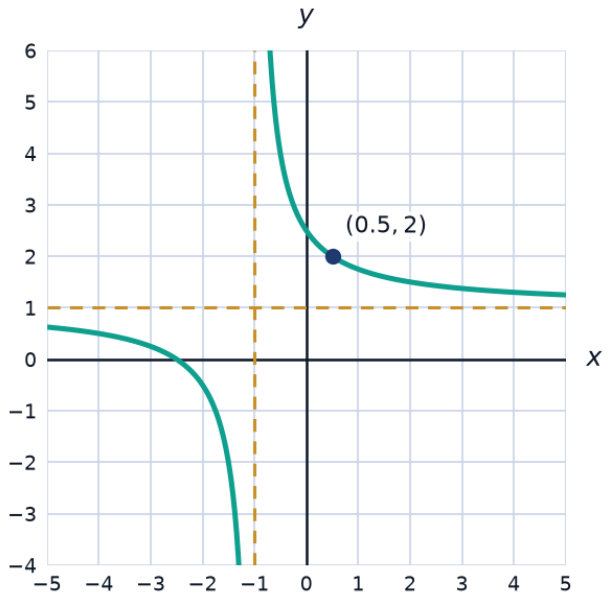
Evaluate $\lim_{x \to 3} \sqrt{x + 13}$.

9 **[Intermediate]** • **Target: Evaluating limits by factoring (the 0/0 form)**

Evaluate $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$.

10 **[Intermediate]** • **Target: Evaluating limits by factoring (the 0/0 form)**

Evaluate $\lim_{x \to -2} \frac{x^2 + 5x + 6}{x + 2}$.



11 **[Intermediate]** • **Target: Evaluating limits by factoring (the 0/0 form)**

Evaluate $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$.

12 **[Intermediate]** • **Target: Evaluating limits by factoring (the 0/0 form)**

Evaluate $\lim_{x \to 1} \frac{x^2 - 1}{x^2 - 3x + 2}$.

13 **[Intermediate]** • **Target: Limits at infinity**

Evaluate $\lim_{x \to \infty} \frac{3x^2 + 2x}{x^2 - 5}$.

14 **[Intermediate]** • **Target: Limits at infinity**

Evaluate $\lim_{x \to \infty} \frac{2x + 7}{x^2 + 1}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Limits at infinity • Cross-Lesson: 2.01 Functions & Transformations

Evaluate $\lim_{x\to\infty} \frac{5x^3-2}{x^3+4x}$.

16 [Challenging] • Target: One-sided limits • Cross-Lesson: 3.01 Sequences & Series

A function is defined by $f(x) = x + 1$ for $x < 2$, and $f(x) = 2x - 1$ for $x \geq 2$. Find $\lim_{x\to 2^-} f(x)$.

17 [Challenging] • Target: One-sided limits • Cross-Lesson: 4.01 Trigonometry

Using the same piecewise function, find $\lim_{x\to 2^+} f(x)$.

18 [Challenging] • Target: One-sided limits • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Using the same piecewise function, does $\lim_{x\to 2} f(x)$ exist? State its value.

19 [Challenging] • Target: The power rule • Cross-Lesson: 7.01 Integration & Applications

Differentiate $f(x) = x^5$.

20 [Challenging] • Target: The power rule • Cross-Lesson: 8.01 Exponentials & Logarithms

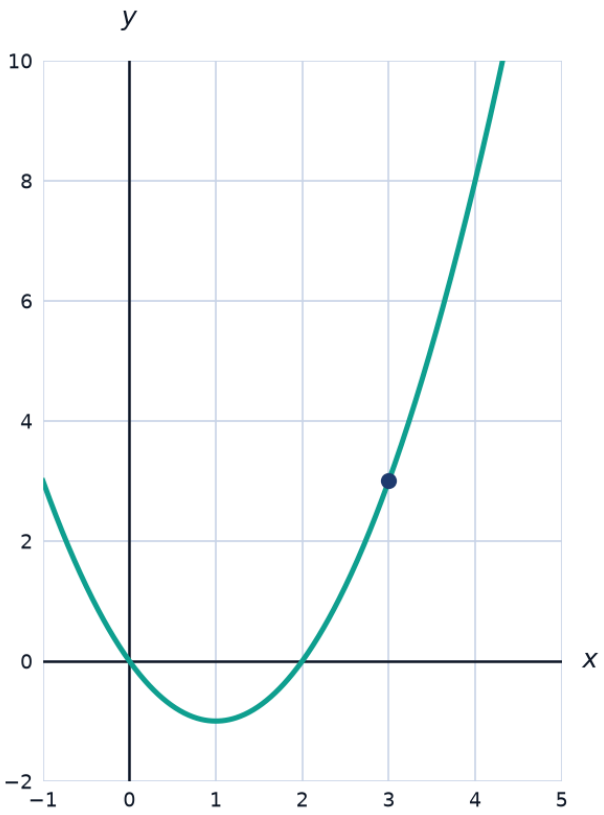
Differentiate $f(x) = 3x^4 - 2x^2 + 7$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: MCQ — Tangent lines

The graph of $f(x) = x^2 - 2x$ is shown, with the point at $x = 3$ marked. Find the equation of the tangent line there.



a. $y = 3x - 9$

b. $y = 4x - 3$

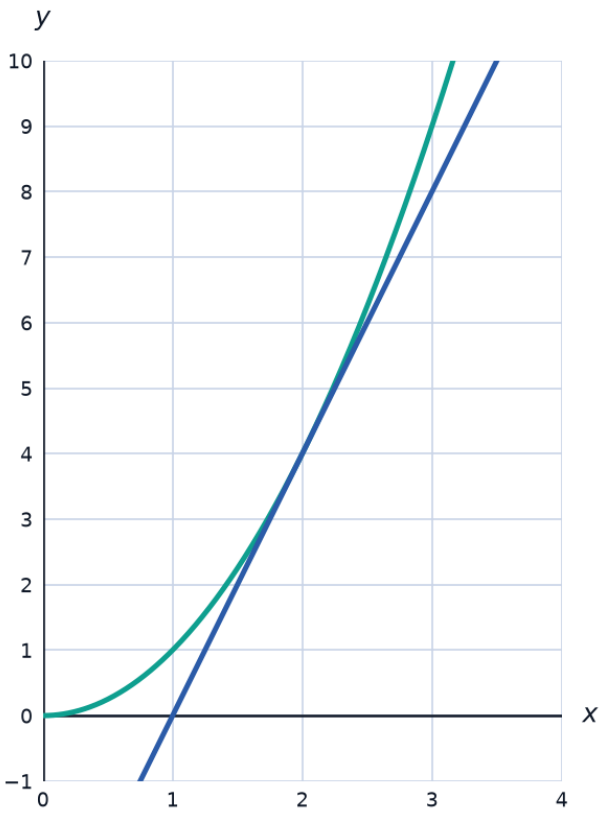
c. $y = 4x + 9$

d. $y = 4x - 9$

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22 [Foundational] • Target: MCQ — Tangent lines

The graph shows $f(x) = x^2$ and its tangent line at $x = 2$. Find the slope of the tangent line by reading the graph.



a. 4

b. 2

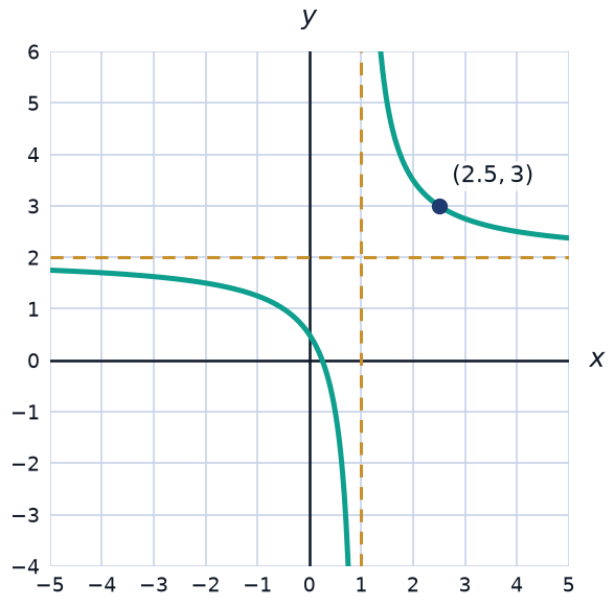
c. 1

d. 8

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate $\lim_{x \to 3} (2x^2 - 5x + 3)$.



a. 12

b. -6

c. 18

d. 6

24 [Foundational] • Target: MCQ — Evaluating limits by direct substitution

Evaluate $\lim_{x \to 3} (2x^2 - 5x + 3)$.

a. 12

b. -6

c. 18

d. 6

25 [Foundational] • Target: MCQ — Evaluating limits by direct substitution

Evaluate $\lim_{x \to 4} \frac{x^2 - 3x}{2x + 1}$.

a. $\frac{4}{7}$

b. $\frac{16}{9}$

c. $\frac{4}{8}$

d. $\frac{4}{9}$

26 [Foundational] • Target: MCQ — Limits at infinity

Evaluate $\lim_{x \to \infty} \frac{4x^2 + 3x}{2x^2 - 7}$.

a. 2

b. 0

c. $\frac{3}{7}$

d. 4

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27 [Foundational] • Target: MCQ — One-sided limits

A function is defined by $f(x) = x + 2$ for $x < 3$, and $f(x) = 3x - 4$ for $x \geq 3$. Find $\lim_{x \to 3^-} f(x)$.

a. 3

b. 4

c. 9

d. 5

28 [Foundational] • Target: MCQ — The power rule

Differentiate $f(x) = x^7$.

a. x^6

b. $7x^8$

c. $6x^7$

d. $7x^6$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — The product rule

Differentiate $f(x) = x^3(2x + 5)$ using the product rule.

a. $8x^3 + 15x^2$

b. $6x^2 + 5$

c. $8x^3 + 5x^2$

d. $2x^3 + 15x^2$

30 [Intermediate] • Target: MCQ — The chain rule

Differentiate $f(x) = (2x + 3)^5$.

a. $5(2x + 3)^4$

b. $2(2x + 3)^4$

c. $10(2x + 3)^5$

d. $10(2x + 3)^4$

31 [Intermediate] • Target: MCQ — The chain rule

Differentiate $f(x) = (x^3 + 2)^2$.

a. $6x^2(x^3 + 2)$

b. $6x^2(x^3 + 2)^2$

c. $3x^2(x^3 + 2)$

d. $2(x^3 + 2)$

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32 [Intermediate] • Target: MCQ — Tangent lines

Find the slope of the tangent line to $f(x) = x^2$ at $x = 4$.

a. 4

b. 2

c. 16

d. 8

33 [Intermediate] • Target: MCQ — Increasing, decreasing, and the second derivative

For $f(x) = -x^2 + 8x$, determine the interval where f is decreasing.

a. $x < 4$

b. $x > 8$

c. $x > 4$

d. $x < 8$

34 [Intermediate] • Target: MCQ — Word problems: rates of change

The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the time at which the ball reaches its maximum height.

a. $t = 3$

b. $t = 5$

c. $t = 6$

d. $t = 2$

35 [Intermediate] • Target: MCQ — One-sided limits

Evaluate $\lim_{x \to -1} (2x^2 + 8x - 9)$.

a. -18

b. 15

c. -12

d. -15

36 [Intermediate] • Target: MCQ — The power rule

Differentiate $f(x) = x^4 - 2x$.

a. $5x^3 - 2$

b. $4x^3 - 2$

c. $4x^4 - 2$

d. $x^4 - 2$

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37 [Intermediate] • Target: MCQ — The product rule

Differentiate $f(x) = 3\cos x$.

a. $-3\sin x$

b. $3\cos x$

c. $3\sin x$

d. $4 - \sin x$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — The chain rule • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Evaluate $\lim_{x \to -1} \frac{x^2 - 2x - 3}{x + 1}$.

a. 4

b. -4

c. -6

d. -2

39 [Challenging] • Target: MCQ — Derivatives of trigonometric and exponential functions • Cross-Lesson: 7.01 Integration & Applications

Differentiate $f(x) = (3x + 6)(-2x - 6)$ using the product rule.

a. $-12x - 30$

b. $-12x + 30$

c. $-12x - 28$

d. $12x - 30$

40 [Challenging] • Target: MCQ — Tangent lines • Cross-Lesson: 8.01 Exponentials & Logarithms

Differentiate $f(x) = (-3x - 4)^5$.

a. $-18(-3x - 4)^4$

b. $-15(-3x - 4)^5$

c. $-15(-3x - 4)^4$

d. $5(-3x - 4)^4$

41 [Challenging] • Target: MCQ — Increasing, decreasing, and the second derivative • Cross-Lesson: 9.01 Statistics & Probability

Evaluate $\lim_{x \to \infty} \frac{4x^2 + 8x}{3x^2 - 8}$.

a. 0

b. $\frac{7}{3}$

c. $\frac{3}{4}$

d. $\frac{4}{3}$

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42 [Challenging] • Target: MCQ — Word problems: rates of change • Cross-Lesson: 1.01 Algebra & Equations

Find the equation of the tangent line to $f(x) = 2x^2 + 3x - 1$ at $x = -3$.

a. $y = 9x - 19$

b. $y = -9x + 19$

c. $y = -9x - 16$

d. $y = -9x - 19$

43 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Functions & Transformations

For $f(x) = -x^2 - 3$, determine the interval where f is increasing.

a. $x < 0$

b. $x > 2$

c. $x < 2$

d. $x > 0$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Sequences & Series

Evaluate $\lim_{x \to 4} \frac{x^2 - 3x}{2x + 1}$.

a. $\frac{4}{7}$

b. $\frac{16}{9}$

c. $\frac{4}{8}$

d. $\frac{4}{9}$

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Trigonometry

Evaluate $\lim_{x \to 5} \frac{x^2 + x - 30}{x - 5}$.

a. 11

b. 13

c. -11

d. 9

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